

When Does Independent Audit Help Agentic Research?

CROSSAUDIT and a Causal Evaluation Framework

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Pre-results draft. The protocol, software, sealed six-task feasibility cohort, and prospective three-vendor design are complete. The 150–180-task human-adjudicated study has not run. No sentence below treats feasibility or power simulation as efficacy.

Abstract

Language-model agents increasingly generate scientific code, analyses, and manuscripts, while their review is commonly performed by the same model or platform. A different vendor may remove exact self-evaluation, but vendor heterogeneity is neither necessary nor sufficient for independent error. We introduce CROSSAUDIT, a git-native protocol that makes supervision inspectable: versioned rules, deterministic checks, bounded model revision, fail-closed admission, and a ledger of findings and dispositions. We formulate the target as net correction under false-block, revision-harm, model-cost, and human-attention constraints. A sealed 542-call feasibility cohort on six deterministic tasks exercised seven measurement paths but produced direction-specific cross-auditor effects, a circular deterministic ceiling, and no structured-ledger advantage over an ordinary log on proxy decisions. It does not establish efficacy. Those failures motivate a prospective study with 150–180 tasks, three vendors, six pinned models, same-model and same-vendor/ different-model controls, three domains, human-blinded gold, and task-clustered inference. CROSSAUDIT contributes an executable protocol and a falsifiable evaluation of when heterogeneous oversight helps; the positive efficacy claim is deliberately reserved for the unrun confirmatory study.

1 Introduction

Agentic research systems can propose hypotheses, execute code, search literature, and draft papers with limited human intervention [1, 2, 3]. Their scale changes the supervision problem. A human can no longer reread every intermediate artefact, yet an automated reviewer may share the Generator’s blind spots. LLM judges exhibit position, style, and self-preference effects [5, 4]; reliable repetition need not imply valid judgement [6]. At the same time, the review process is often inaccessible outside a platform.

The tempting remedy is “use another vendor.” That intervention removes exact self-grading, but it bundles several mechanisms: model capability, training and post-training, prompting infrastructure, data overlap, and stochasticity. Cross-vendor is therefore an assignment label, not proof of independence. The scientific question is narrower and testable:

Under fixed information, budget, and controller rules, when does a more heterogeneous Auditor improve final artefact quality without unacceptable clean-artifact burden or revision harm?

We make four contributions. First, we separate same-model, same-vendor/ different-model, and cross-vendor review. Second, we specify CROSSAUDIT, an executable supervision protocol whose decisions become inspectable artefacts. Third, we define a causal estimand and operational utility that include false blocks, new defects, unnecessary changes, model cost, and human attention. Fourth, we release a prospective three-vendor confirmatory package: allocation, human adjudication, power simulation, analysis plan, refusal preflight, and tests. We also report the sealed feasibility cohort that shaped the design, including its negative boundaries.

The current evidence establishes implementation and measurement feasibility, not that cross-vendor audit improves science. This distinction is the main organising choice of the paper.

Table 1: The eight protocol invariants and their empirical role.

Invariant	Requirement	Failure it makes measurable
I1	declared cross-vendor production audit	exact self-evaluation; not all correlated error
I2	append-only parsed ledger	omitted or overwritten findings
I3	schema and rule-ID validity	confident but ungrounded verdicts
I4	deterministic precedence	model arithmetic or waiver of hard checks
I5	bounded rounds	oscillation and unbounded cost
I6	graded escalation	silent failure versus human interruption
I7	receipt binding	stale rules, prompts, checks, or artefacts
I8	fail-closed admission	audit theatre that refuses no action

2 From same-source concern to a causal question

Self-preference is documented for some LLM evaluators judging their own generations [4], but later work shows that quality and identity blinding can explain part of the measured effect [7, 8]. Panel-of-LLM evaluation uses model diversity to dilute idiosyncratic bias [9], yet it does not isolate vendor from model ability. Scientific audit adds asymmetric costs absent from ordinary ranking: a false block consumes human attention, and a revision can introduce a new defect.

Let E_G denote a Generator defect and E_A failure by an Auditor to block it. The relevant association is

$$\rho_{GA} = \text{Corr}(E_G, E_A \mid X, M_G, M_A, H), \quad (1)$$

where X is the task, M_G and M_A are resolved model configurations, and H fixes the harness, prompt, information, and controller. Vendor does not appear in the definition. It is an assignment attribute that may change ρ_{GA} , auditor capability, both, or neither.

For supervision condition s , define

$$U(s) = v_r P_s(R) - c_b P_s(B_0) - c_n P_s(N) - c_u P_s(U) - c_m M_s - c_h H_s, \quad (2)$$

where R is successful repair, B_0 an unnecessary block or escalation of a clean artefact, N a revision-introduced defect, U an unnecessary change, M_s model cost, and H_s human attention. The weights are deployment-specific. Unless frozen before results, we report the components and their Pareto frontier instead of choosing weights post hoc.

This yields four predictions. (i) Same-model, same-vendor/different-model, and cross-vendor review need not be monotonically ordered. (ii) Deterministic checks should dominate on decidable defects, but circular evaluation occurs if the same checker defines gold. (iii) broad rules may increase recall and clean false blocks together. (iv) hard gates may improve repair while increasing cost, unnecessary changes, or rule-surface adaptation.

3 The CROSSAUDIT protocol

Each research increment is a pinned commit in a Science Repository. A Generator can write that repository but not the Audit Repository. An Auditor receives the pinned artefact as fenced data, a versioned Constitution, and no Generator transcript. Deterministic checks run first. The Auditor returns a schema-valid verdict and rule-cited findings. The controller permits bounded revision, commits each report and disposition, and escalates unresolved or invalid states. A verified receipt binds the increment, rulebook, checks, prompt, and decision. Protected admission should accept only a verified PASS.

The present reference implementation does not yet provide the strongest form of all invariants. It preserves parsed reports rather than raw model exchanges; vendor identity can be self-declared; the operator can rewrite unpublished git history; and binding admission to platform branch protection remains a deployer action. Thus “third-party-auditable” presently means internally consistent as published, not externally anchored or cryptographically attested.

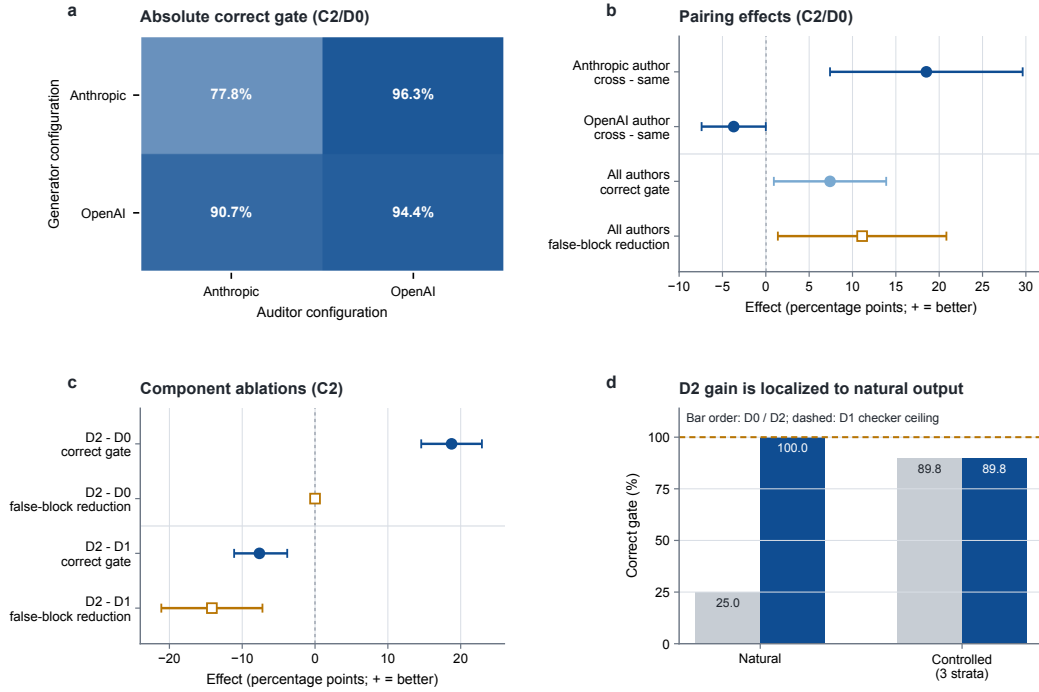


Figure 1: Sealed six-task feasibility results. Cross-minus-same correct gating was positive on average but reversed by Generator direction. Combined DCL plus model improved over model-only only on natural outputs and remained below the circular DCL ceiling. Constitution effects are not estimated because the frozen scorer used repeat 0 where the design called for three repeats.

4 What the sealed feasibility cohort established

The v4 feasibility cohort used two model configurations and six deterministic convenience tasks. It scheduled 542 calls across Generator–Auditor assignment, DCL/Constitution ablations, defensive-production policies, evidence strata, whole-loop revision, ledger surfaces, and three-repeat integrity cells. An outcome-free seal preceded scoring. All values below are descriptive and configuration-specific; the independent sample is six tasks, not 542 calls.

The pooled cross-minus-same correct-gate contrast was +7.4 percentage points and the clean false-block contrast −11.1 points. The pooled value concealed directional changes of +18.5 and −3.7 points. Combined DCL/model gating was +18.8 points above model-only, entirely on natural outputs, but −7.6 points below DCL-only. The latter checker also defined the micro-task gold labels and is a harness ceiling, not independent validation.

Across 24 seeded whole-loop branches, same and cross conditions each resolved 12/12 initial defects and ended with 11/12 acceptable artefacts, 1/12 new defects, and 3/12 unnecessary changes. On seven deterministic episodes, the structured ledger matched the ordinary log on accept and tamper proxy accuracy (71.4% each) and was worse on first-defective-round reconstruction (57.1% versus 92.9%). Proxy reviewers were models, not humans. Hard gating repaired two of nine initially wrong text outputs without regressing three initially correct ones, but required more than twice the calls and cost; a two-task code subset was ceiling-limited and showed compliance-only changes rather than measured quality gain.

These findings establish that the harness can execute, stop, seal, and measure. They reject a universal reading of “cross-vendor is better” and motivated the design below. They are not pooled with the confirmatory study, used for observed power, or promoted to a vendor-population claim.

Table 2: Minimum prospective design and refusal gates.

Element	Requirement
Tasks	150, three domains, ≥ 60 real replays; blind expansion ≤ 180
Models	3 vendors, 6 resolved snapshots; complete 3×3 primary matrix
Controls	same snapshot, same-vendor/different-model, cross-vendor
Human gold	two blinded domain reviewers plus third adjudicator; separate Gold and Matching Panels
Calibration	30 independent items; Krippendorff $\alpha \geq 0.80$, raw agreement ≥ 0.85
Primary gate	correct-gate superiority and clean-burden +5-point non-inferiority
Integrity	provider-returned identity, frozen prompts/tasks/code, external times-tamp, outcome-blind 12-task dry run
Scope	included configurations only; every direction and leave-one-vendor-out mandatory

5 Prospective confirmatory evaluation

The v5 package preserves v4 unchanged and begins a new outcome-free design. The initial sample is 150 independent task briefs, equally blocked over three domains, with one blinded expansion to at most 180. At least 60 are privacy- approved real-task or repository replays. Three principal models, one from each vendor, form a complete Generator $\times$ Auditor matrix. Three alternate auditor snapshots supply the missing same-vendor/different-model control.

Each task supplies natural principal-model output and, where applicable, a verified-clean/exactly-one-mutant sibling pair. Unusual but correct artefacts are hard negatives. Every required cell uses three fresh-context repeats. Constitution levels C0/C1/C2 each receive three repeats on a balanced 60-task controlled subset, removing the v4 estimand mismatch. DCL-only gold is created by independent human or held-out deterministic validation, never by the tested checker itself.

5.1 Estimand and inference

After collapsing repeats and fixed target strata, task t contributes

$$C_t = \frac{1}{3} \sum_g \left[\frac{1}{2} \sum_{a \neq g} Y_{tga} - Y_{tgg} \right]. \quad (3)$$

The primary point estimate is the mean of C_t . Whole-task bootstrap-t intervals preserve artefacts, siblings, defects, and calls inside the independent unit. A crossed logistic mixed-effects model with task and base- artefact intercepts and fixed Generator, Auditor, separation, domain, stratum, and defect-channel terms is a sensitivity analysis. Vendor is fixed; three vendors cannot identify a random vendor-population effect.

The headline requires both (i) the lower endpoint of a two-sided 95% interval for cross-minus-matched `correct_gate` above zero and (ii) the upper endpoint of a one-sided 95% interval for clean unnecessary escalation below +0.05. Invalid, missing, or identity-unverified assigned audits are incorrect gates and clean unnecessary escalations in the service-level intention-to-audit population. Direction-specific, S1-versus-S0, review-only, DCL-negative, cost/latency/human-minute, repeat-flip, and leave-one-vendor-out results are mandatory.

5.2 Power, cost, and go/no-go

The simulator mirrors three Generators, natural/clean/mutant artefacts, the complete 3×3 audit matrix, alternate same-vendor Auditors, task and base-artefact dependence, and three correlated repeats. It uses no observed v4 outcome. Before dispatch, at least 10,000 datasets per frozen scenario must show $\geq 80\%$ conjunctive power at an eight-point effect, calibrated boundary type-I error, and 93–97%

interval coverage. $N=150$ failure permits one prospective increase to 180; $N=180$ failure converts the paper to estimation or stops dispatch.

A 10,000-dataset central planning simulation that read no observed outcomes returned a 7.998-point Monte Carlo marginal effect for the nominal eight-point scenario, 99.65% conjunctive power, 94.87% interval coverage, 2.49% directional superiority type-I error, and 4.75% non-inferiority-boundary type-I error. This validates the central allocation under the current normal task-level design interval. The frozen stress scenarios and calibration against the exact bootstrap-t analysis remain dispatch blockers.

The planned ceiling is 24,930 model calls before separately authorised technical retries: 450 generations, 12,150 primary audits, 4,050 same-vendor alternate audits, 6,480 Constitution additions, and about 1,800 whole-loop calls. The executable preflight refuses dispatch until money, tokens, wall time, human hours, privacy approval, identity locks, panel calibration, power, randomisation, and the dry-run receipt are filled and hash-bound. A call ceiling is not permission to spend.

6 Defensive adaptation and ledger utility

Hard rules can improve work or merely reshape it. The confirmatory whole-loop subset therefore reports final acceptability, resolved defects, new defects, unnecessary changes, and human minutes rather than treating more assertions, documentation, or rule citations as quality. Policy P0 hides audit, P1 discloses non-gating shadow audit, and P2 supplies hard-gate findings under the same maximum rounds. Evidence for harmful defensive programming requires increased compliance-only or defensive content without improved blinded final quality or held-out performance. Verbosity alone is not evidence.

Ledger utility is similarly comparative. Human reviewers, not model proxies, are crossed with episodes under final-only (E0), ordinary transcript (E1), and structured ledger (E2) surfaces. Primary outcomes are accept/reject accuracy and time to a correct decision; tamper detection and first-defective-round reconstruction are secondary. E2 must beat E1, not merely E0, to justify the structured representation. Unless separately powered and frozen, this module remains an appendix and cannot rescue the primary audit result.

7 Limitations and responsible scope

The protocol audits the research record, not the world. It cannot detect a miscalibrated instrument, a wrong but accurately described dataset, or fraud upstream of the repository. Models may share public training corpora across vendors. A malicious operator can omit failed private runs or rewrite history before publication. Prompt injection can shape the stochastic report even when the Auditor is toolless and the DCL is non-overridable. Shipping unpublished research to another provider creates confidentiality and intellectual-property obligations outside the protocol.

The confirmatory study, even if positive, supports only its included models, vendors, domains, task frame, prompts, and controller. Model snapshots age. Real-task replay can be selected unrepresentatively. Human gold can be wrong; the calibration gate controls but does not eliminate that risk. The utility weights in Eq. 2 are not universal, so a deployment with costly false blocks may rationally reject a configuration that improves average gate accuracy.

8 Conclusion

CROSSAUDIT makes automated supervision inspectable and falsifiable: deterministic precedence, rule-cited judgement, bounded revision, receipts, and a versioned ledger. The sealed feasibility cohort demonstrates the machinery and exposes why a second vendor cannot be assumed to be a better judge. The prospective study separates model change from vendor change and evaluates final net correction under clean-burden and cost constraints. Until it runs, the evidence supports a protocol and

an executable evaluation framework, not efficacy. An AI scientist should not grade its own homework; this work specifies how to test whether a different grader actually helps.

Broader impact. Inspectable audit could reduce silent errors and make machine-assisted research easier to scrutinise. It could also create false confidence, centralise rule-making, leak confidential work to providers, or encourage optimisation to a narrow Constitution. We therefore require explicit claim scope, human authority over rules and escalations, privacy approval, negative-result reporting, and visible distinction between measured zero and unmeasured state.

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